Cyber Security and Privacy
MS6880

Information Privacy: Economics and Strategy

Saji K Mathew, PhD
Professor, Management Studies
INDIAN INTITUTE OF TECHNOLOGY MADRAS

MY DAD TOLD ME YOU'RE SPYING ON US
HE'S NOT YOUR DAD
imgflip.com

Valuation of privacy

- The protection and disclosure of personal data are likely to generate trade-offs with tangible economic dimensions.
- Understanding the value that individuals assign to the protection of their personal data is of great importance to policy makers, businesses, and researchers.
  - Example: how much must be invested in data security

NPTEL

Another example: Cloud contracts

- Legal complexities as the servers can be located in more than one geographical location(Pearson and Charlesworth, 2009)
- Cloud providers are largely concerned with providing service and additional revenue generating mechanisms such as providing their data to advertisers (Pfarr et al. 2014).
- Users of cloud do not possess the technical mechanisms that will allow them to control the usage of their data which leads to significant risk to data.
- Cloud contracts as a way to manage risk (Pearson and Benameur, 2010)
- Liabilities are not very strong as providers exclude or restrict liability as much as possible (Hon et al., 2012).

NPTEL

Privacy preference inconsistency

- *Maximum* price a person would be willing to pay to acquire a good she did not own (WTP), and the *lowest* price she would be willing to accept to part with the same good if she initially owned it (WTA).
- Loss aversion built in human response
- Endowment effect stems from parting pain
- WTA tends to produce valuations that are substantially higher than those produced by WTP

NPTEL

NPTEL

The Dark Side of C[redacted]

The stake holders

- IFA
  - Life & Health insurance
  - Piloted Smart Choice
- ShopSense
  - Grocery chain
  - Analytics driven
  - Sells customer data
- Customers
  - Insurance
  - grocery

Consequence analysis

- IFA
  - Accurate risk leads to differentiated premiums (+)
  - Data could give competitive advantage (+)
  - Pilot successful (+)
  - Data could be unreliable (-)
  - Loss of customer goodwill (-)
  - Dependence on ShopSense (-)
- ShopSense
  - Direct revenues(+)
  - No extra costs (+)
  - Erosion of loyalty(-)
  - Loss of customer goodwill(-)
  - Difficulty of monitoring (-)

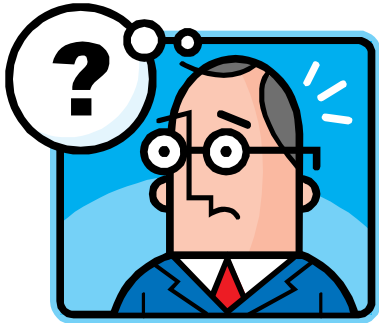
NPTEL

The victim: customer

- Intimacy without consent!
- Customer pays for low reliability of model
- Customer concerns not represented

- How to enhance customer trust?

NPTEL

Solutions?

- Joint offering of insurance product